

Yashas D

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SUMMARY

Artificial Intelligence & Machine Learning professional with hands-on experience building LLM-based Industrial Copilots, Generative AI systems, and automation solutions. Passionate about contributing to innovative, scalable AI projects while expanding expertise in intelligent systems and automation..

SKILLS

Programming Languages: Python, C#
Machine Learning & AI: TensorFlow, CNNs, Pandas, NumPy, OpenCV, MediaPipe, FastAPI
Generative AI: LLMs, Prompt Engineering, AI Agents, RAG Pipelines, NLP
Data: MySQL, Azure Blob Storage, Apache Jena, openpyxl
Tools & Platforms: Docker, Grafana, Promptfoo, OpenAI APIs, Power BI, GitHub
AI Evaluation: LLM Evaluation, Prompt Analysis, Agentic Workflow Automation
Soft Skills: Communication, Critical Thinking, Problem Solving, Teamwork

EXPERIENCE

Graduate Trainee Engineer - SIEMENS Technologies & Services Pvt Ltd Aug 2025 - Present

Project: Industrial Copilot for Engineering (ICFE) - TIA Portal

- **Tech Stack:** C# | Microsoft Semantic Kernel | GIT | Docker | Azure | Kernel Functions | LLMs
- Engineered AI-driven PLC Code Agent features and project context plugins within the TIA Portal Copilot platform, implementing scalable multi-agent workflows, kernel functions, and LLM-based automation for engineering task execution.
- **PLC Code Agent Development:** Designed and implemented scalable PLC Code Agent features and plugins by analyzing project requirements, defining technical specifications, and integrating orchestrator-based workflows that process user prompts, route requests to specialized agents.
- **Feature Development & Refactoring:** Developed new functionalities and performed code refactoring to improve the performance, reliability, and maintainability of the PLC Code Agent.
- **LLM Evaluation & Monitoring:** Conducted prompt evaluation using Promptfoo and monitored system metrics through Grafana dashboards to ensure stable and reliable LLM responses.
- **Cloud Deployment & CI/CD:** Deployed the application on Azure for production use and managed CI/CD pipelines via GIT, ensuring scalable, reliable, and streamlined application delivery.

Generative AI Intern -YOKOGAWA India

Sept 2024 - Jul 2025

- **AI-Powered Document Processing:** Developed OCR pipelines using Tesseract, PyMuPDF, and PyPDF2 to extract data from unstructured PDFs and convert it into structured Excel reports using pandas and openpyxl addressing real-world organizational challenges.
- **Backend Automation & APIs:** Built scalable FastAPI-based APIs to automate data extraction, validation, and reconciliation workflows across enterprise systems.
- **Generative AI Integration:** Implemented GPT-based document analysis and comparison systems for summarization, translation, and intelligent document insights for Y-GPT.
- **Enterprise Automation Systems:** Designed and implemented EAS for GSTR-3B reconciliation, invoice validation, and dispatch document processing, integrating Azure Blob Storage and SQL databases for scalable processing, which reduced manual data entry by 60%.
- **AI-Driven Document Management:** Developed solutions for document comparison, compression, and version tracking, reducing manual review effort by 80% and improving compliance workflows.

RESEARCH & INNOVATION

- Author of **8 IEEE research publications** in Artificial Intelligence and Machine Learning domains, indexed in Scopus.
- Presented 1 research paper in a **Springer conference**.
- Inventor on **2 patents filed under the Indian Patents Act, 1970** related to AI-driven systems.
- Participated in **25+ hackathons**, securing multiple top rankings and national-level recognitions.

PROJECTS

Repo Analyzer

Developed an AI-powered repository analysis platform that analyzes GitHub repositories, extracts project structures, and generates intelligent code summaries using Large Language Models. Built scalable FastAPI-based backend services for repository parsing, metadata extraction, dependency analysis, and automated documentation workflows using Python, FastAPI, LLMs, NLP, and GitHub APIs. (Feb 2025)

Gesture-Controlled AI for Seamless Digital Interaction

Developed a real-time gesture recognition platform integrating a virtual mouse, AI voice assistant, virtual keyboard, calculator, and drawing canvas into a unified interaction system. Implemented computer vision-based hand tracking and motion-sensing workflows using Generative AI, MediaPipe, and OpenCV for touchless digital interaction and enhanced accessibility. (Oct 2024)

AI-Powered Healthcare Chatbot System

Developed an AI-powered healthcare chatbot capable of providing symptom-based guidance, health recommendations, and intelligent conversational support using NLP and Machine Learning techniques. Built backend workflows for automated response generation, healthcare data handling, and real-time interaction management to improve healthcare accessibility and preliminary health assistance. (Jul 2024)

EDUCATION

2021 - 2025 B.E (Hons) - Artificial Intelligence & Machine Learning at NHCE, Bangalore (CGPA: 9.38)
2019 - 2021 Presidency Pre-University College, Sira, Tumkur, Karnataka (Percentage: 91.5)

CERTIFICATES

- **Microsoft Azure Machine Learning** — Microsoft
- **Generative AI with Large Language Models** — DeepLearning.AI & AWS
- **Retrieval Augmented Generation (RAG)** - DeepLearning.AI

PARTICIPATIONS

- **Runner-Up in Hack Kshetra** organized by **VVCE, Mysuru** - Oct, 2024
- **Ranked 6th** among 5000+ students in **TechThon** by **Visionet Systems**, 2024
- **Top 17th** among the 2,600+ teams in **TATA INNOVENT** by **TATA Technologies**, 2024
- **Top 25th** among the 50,000+ participants **Vultr Cloud Hackathon** organized by **Vultr**, 2024

EXTRA-CURRICULAR ACTIVITIES

- **Vice President**, Green Warriors Club, NHCE (Sept 2023 - Jul 2025)
- Volunteered in lake restoration and water conservation drives through the Jalposhan Club